

Constraint-Aware Candidate-Set MAPPO for Stable Routing in Dynamic LEO Networks

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Abstract-- Low-Earth-orbit routing must react to topology and congestion without replacing a still-feasible cached next hop unnecessarily. We formulate decision-level avoidable switching as an explicit constrained objective for distributed candidate-set MAPPO. A permutation-equivariant actor scores feasible neighbors; a graph critic and centered local credit are used only during centralized training. The controller applies a projected Lagrange multiplier to a differentiable conditional-switch surrogate. In a preregistered sealed test covering two dynamic 24-satellite scenarios, eight independently trained policy seeds, and 50 paired unseen workloads, the constrained policy reduced avoidable-switch rates by 15.22 and 36.25 percentage points. Delivery effects were -0.96 and $+0.77$ points and met a prospective two-point non-inferiority margin. Every integrity, budget, reduction, and delivery gate passed. The claim is limited to the audited simulator and these two traffic regimes.

Index Terms-- LEO satellite networks, multi-agent reinforcement learning, MAPPO, routing stability, constrained optimization.

I. Introduction

Low-Earth-orbit (LEO) constellations create a routing problem in which connectivity changes predictably with orbital motion while queues and traffic concentrations vary on shorter time scales. A next hop that is geometrically attractive can become congested, and a route that was recently selected can disappear. Classical shortest-path recomputation handles graph change, but frequent recomputation or local policy oscillation can also increase forwarding-state churn. A learning policy optimized only for delivery has no reason to retain a usable cached choice when another candidate has a marginally higher instantaneous score.

We focus on a narrow operational event: an *avoidable switch* at a local forwarding decision. It occurs only when the packet's cached next hop remains feasible, at least one alternative is feasible, and the policy selects a different neighbor. A first decision and a change forced by link infeasibility are not counted. The definition separates policy-induced instability from movement-induced rerouting and yields an integer numerator and denominator that can be audited before contention changes the physical outcome.

The central question is not whether MAPPO is globally superior to classical routing. It is whether an explicit decision-level constraint can reduce unnecessary next-hop changes while preserving delivery within a tolerance fixed before test access. This distinction matters: a reward penalty mixes the units of stability with the rest of the QoS objective, whereas a rate budget states the operational target directly.

We use centralized training with decentralized execution (CTDE). Each satellite applies the same candidate-set actor using only its packet, cache, incident-link, and one-hop congestion context. During training, a topology-aware critic sees the full graph state. The actor remains valid when candidate order changes because the same encoder and scorer are applied to every neighbor and the candidate context is pooled symmetrically.

Our contributions are threefold. First, we define and instrument a pre-contention decision-level stability endpoint that excludes forced changes. Second, we add a projected dual controller to candidate-set MAPPO and use a partition-stable conditional-probability surrogate. Third, we evaluate the method under a one-time, hash-bound test protocol in which policy seed is the independent replication unit and 50 workloads are paired common random numbers. MAPPO,

PPO, set encoders, and graph critics are established components; novelty is claimed only for their constraint formulation and audited evaluation in this setting.

II. Related Work

Q-routing learns local delay estimates from downstream feedback and remains a natural adaptive baseline [1]. Hypatia demonstrates constellation-scale packet experimentation over time-varying satellite connectivity [2]. Our simulator is smaller and synthetic, so it supports controlled causal comparison rather than orbital-scale realism. Deep Sets provides the invariance and equivariance principles behind shared candidate scoring [3]. PPO supplies the clipped policy update [4], and MAPPO shows that centralized value learning can be effective in cooperative multi-agent tasks [5].

Graph attention networks motivate topology-conditioned aggregation in the training-only critic [6]. Reproducibility work in deep reinforcement learning emphasizes independent runs, complete reporting, and sensitivity to evaluation design [7], [8]. We operationalize those concerns by preassigning policy seeds, separating checkpoint selection from validation and test workloads, using seed-level inference, and retaining failed pre-test executions rather than silently replacing them.

CTDE also connects to actor-critic methods such as MADDPG and counterfactual policy gradients [9], [10]. Our shared actor differs in that all satellites use one permutation-equivariant candidate policy, while the centralized component supplies only a value estimate. Constrained policy optimization targets monotonic constrained improvement [11]; our projected PPO Lagrangian is deliberately simpler and makes no comparable theoretical guarantee. Safety Gym illustrates the importance of reporting both task return and constraint cost [12].

For LEO routing, propagation-aware path selection and constellation design expose trade-offs not present in terrestrial graphs [13], [14]. Surveys of small-satellite networking further emphasize mobility, link intermittency, and protocol integration [15]. Those works motivate external replication, but they do not turn the current synthetic graph into an orbital-fidelity result. We therefore keep the experimental claim tied to the simulator actually audited.

III. System Model and Objective

A. Dynamic forwarding environment

At slot t , the network is a directed graph $G_t=(V,E_t)$ with $|V|=24$ satellites arranged as four planes of six. Intra-plane and inter-plane links vary with a simplified orbital phase. Each episode lasts 30 synchronous slots. Every satellite with a head-of-line packet proposes one action from its currently feasible neighbors; transmissions are then resolved jointly under per-link service limits. The simulator uses FIFO queues, packet deadlines, a 12-hop limit, and three traffic classes.

The medium-load regime initializes 12 packets and admits six exogenous packets per slot. The hotspot regime initializes 20 and admits 16 per slot. Its destination concentration is stronger, creating sustained contention around a small part of the graph. These are synthetic stress settings: they are not TLE/SGP4 traces and do not reproduce a deployed control plane.

A candidate vector contains 26 normalized fields: current and downstream queue ratios; delay, residual bandwidth, utilization, reliability, and remaining link lifetime; prior contention; destination progress and orbital coordinates; remaining and used hop budgets; a cached-route switch indicator; visited-node and previous-node indicators; topology phase; normalized packet waiting age; and one-hot traffic class. Infeasible actions are removed before sampling. The explicit NO_OP action is available only when the agent has no head-of-line packet.

Group	Candidate fields
Queue	current/downstream ratios; prior contention
Link	delay; bandwidth; load; reliability; lifetime
Geometry	progress; current/candidate/destination orbit
Budget	remaining and used hop ratios
Cache/history	switch; visited; previous-node indicators
Packet/time	waiting age; class one-hot; topology phase

Table I. Grouped 26-dimensional candidate observation.

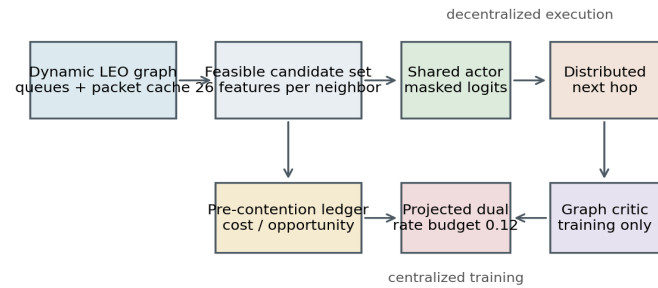


Fig. 1. CTDE pipeline. The critic and dual update are training-only; execution uses local candidate features and a feasibility mask.

B. Decision-level stability ledger

For active decision t , let $o_t=1$ when the cached next hop and at least one alternative are both feasible. Let $c_t=1$ when $o_t=1$ and the selected candidate differs from the cache. Otherwise both indicators are zero as applicable. The rate for policy seed i aggregates integer counts over all paired workloads:

$$R_i = \sum_{j,t} c_{ijt} / \sum_{j,t} o_{ijt}.$$

The ledger is recorded immediately after action selection and before contention resolution. This prevents a proposed avoidable switch from disappearing merely because another transmission wins service. Forced reroutes, inactive agents,

padding, and first-route decisions do not contribute to the numerator. A zero denominator is treated as non-estimable rather than converted to zero.

C. Constrained optimization target

Let $J(\theta)$ denote the expected QoS return under policy parameters θ and let $B=0.12$ be the prespecified avoidable-switch budget. The empirical problem is maximize $J(\theta)$ subject to $E[R(\theta)] \leq B$. We optimize the Lagrangian with a nonnegative multiplier λ . The baseline and constrained arms use exactly the same QoS-only reward, observations, actor, critic, optimizer, and training workloads; only the constraint is enabled for the treatment arm.

The budget is not a hard per-trajectory guarantee. It is a training target combined with prospective test gates on the mean rate, its uncertainty bound, and the maximum of eight policy-seed rates. This layered interpretation prevents a favorable average from masking a single deployed policy identity that clearly violates the operating target.

D. Execution semantics

All active satellites observe the same frozen slot snapshot and act simultaneously. The environment validates joint actions, resolves link service, admits forwarded packets, expires deadlines, adds exogenous arrivals, and advances time in a fixed order. Local execution requires synchronous one-hop queue and contention telemetry. The global state, graph attention messages, bootstrap resampling, and multiplier update are absent from deployment-time action selection.

Because candidate positions have no persistent semantic identity, a flat fixed-slot policy can learn artifacts of enumeration. The shared scorer avoids that dependency: permuting candidate rows permutes output logits, while symmetric pooling leaves the set context unchanged. This property is central when links enter or leave the feasible set.

IV. Constraint-Aware Candidate-Set MAPPO

A. Actor and centralized critic

For each feasible neighbor j , the actor maps feature vector x_j through a shared 32-unit encoder. A masked mean of encoded candidates forms set context z . The shared scorer maps $[h,z]$ to logit l_j ; invalid logits are masked before the categorical distribution is formed. Feasible-action entropy is normalized by the logarithm of the current feasible-action count so that exploration pressure is comparable across changing node degree.

The critic is used only during training. It embeds node and edge fields, applies topology-aware attention, pools graph information, and combines it with global fields to estimate the shared value. Its hidden width is 64. The training reward combines team QoS with centered local credit: subtracting the active-agent mean keeps the local term zero-sum, adding discrimination among simultaneous actions without shifting the team objective.

B. PPO and constraint surrogate

We use generalized advantage estimation with $\gamma=0.99$ and $\lambda_{\text{GAE}}=0.95$. PPO clips the importance ratio at 0.2; advantages and returns are normalized, the entropy coefficient is 0.01, actor and critic gradient norms are clipped at 1 and 10, and target KL is 0.02. Actor and critic use Adam with learning rate $8e-4$ and decay.

At every eligible state, the differentiable stability cost is the policy probability assigned to candidates other than the still-feasible cached next hop. The rollout micro-average weights all eligible decisions equally before minibatch partitioning. If C-hat is that surrogate, the actor loss is

$$L_{\text{actor}} = L_{\text{PPO}} + \lambda_{\text{B}} C\text{-hat}.$$

The constant term $-\lambda_{\text{B}}$ is omitted from actor gradients but retained in the logged centered Lagrangian. One multiplier is fixed across all three epochs and four minibatches from rollout k . After the rollout, the empirical integer rate $R\text{-hat}$ updates

$$\lambda_{k+1} = \text{clip}(\lambda_k + 0.05(R\text{-hat} - 0.12), 0, 5).$$

A rollout without an opportunity skips the dual update. There is no warmup, smoothing, adaptive step size, or hidden controller state. The probability surrogate supplies gradients, while the integer ledger drives the dual and all reported endpoints.

C. Training algorithm

Algorithm 1 Frozen constrained MAPPO update
1: Collect four complete 30-slot episodes with paired workload order.
2: Store masks, log probabilities, rewards, values, and pre-contention c_o .
3: Compute normalized GAE and the rollout conditional-probability surrogate.
4: For three epochs and four shuffled minibatches, update clipped PPO actor plus λ -weighted surrogate; update graph critic.
5: Project actor/critic gradients; stop an epoch at the frozen KL condition.
6: Compute $R\text{-hat} = \text{sum}(c)/\text{sum}(o)$; update λ once when $\text{sum}(o) > 0$.
7: Save periodic checkpoints; continue through the full 50k-step budget.

D. Arms and checkpoint selection

The QoS-only baseline disables the constraint. The constrained arm enables the projected multiplier but retains the same QoS reward. A reward-shaped control retrain the historical policy whose reward includes accepted physical switches. That control is informative but not an equivalent fixed penalty: its event definition is post-contention and therefore differs from the primary decision-level cost.

Every arm and seed trains from initialization. There are 2 scenarios $\times$ 3 arms $\times$ 8 seeds = 48 jobs. Each job completes the first four-episode rollout reaching 50,000 environment steps, yielding an audited boundary of 50,040 steps; performance early stopping is disabled. Periodic checkpoints are written every 5,000 steps and validation occurs every 40 rollouts.

Unconstrained and reward-shaped checkpoints are selected lexicographically by delivery, reward, negative drop rate, and negative mean delay on workloads 77001--77010. The constrained arm first retains candidates with ratio-of-sums rate at most 0.12 and then maximizes delivery, class-2 delivery, reward, drop, delay, rate, and earlier step in a fixed order. All selected checkpoints are reproduced on the selection panel and evaluated separately on workloads 77011--77020 before the sealed panel can be authorized.

Item	Frozen value
Actor / critic width	32 / 64
Optimizer / LR	Adam / $8e-4$ decay
GAE γ / λ	0.99 / 0.95
PPO clip / entropy	0.20 / 0.01
Batch / epochs / minibatches	4 ep. / 3 / 4
Budget / dual step / cap	0.12 / 0.05 / 5
Training horizon	50,000 target steps

Table II. Frozen training and constraint settings.

V. Experimental Methodology

A. Baselines and pairing

The primary reference is the matched QoS-only MAPPO arm because it isolates the effect of adding the decision constraint. Three classical methods broaden context. Q-routing is retrained for each scenario and each of the eight policy identities for 500 episodes with $\alpha=0.3$ and $\epsilon=0.1$ during training. OSPF-ECMP uses current link costs and eight routing-randomness identities. Global Dijkstra is deterministic and uses one sentinel identity. All methods encounter identical physical dynamics and sealed workload seeds.

Information sets are not identical across algorithm families. Dijkstra and OSPF use current link-delay structure; Q-routing adapts delay estimates; MAPPO also observes local queues, packet context, and one-hop contention. The comparison is consequently an end-to-end algorithm-system comparison, not a theorem about equal-information decision rules.

B. Data partition and sealed grid

Workload identities are disjoint by role. Training cycles through 76001--76200. Checkpoint selection uses 77001--77010, and an independent development gate uses 77011--77020. Only after both MAPPO and classical chains were frozen did a separate authorization instantiate test workloads 78001--78050. The complete sealed grid has 4,100 unique rows: 2,400 from three MAPPO arms, 800 Q-routing, 800 OSPF-ECMP, and 100 deterministic Dijkstra rows.

Component	Count / identity
Topology	4 planes $\times$ 6 satellites
Scenarios	medium; hotspot-high
MAPPO arms	baseline; constrained; shaped
Policy seeds	8 independent per arm/scenario
Train / select / gate	200 / 10 / 10 workloads
Sealed workloads	50 paired, used once
Classical methods	Q-routing; OSPF; Dijkstra
Complete test rows	4,100

Table III. Evaluation design.

C. Estimands and replication unit

An episode row is a repeated measurement, not an independent algorithm run. For delivery, we compute the constrained-minus-baseline difference for each paired workload, average within each policy seed, and then give the eight seeds equal weight. For the decision rate, integer costs and opportunities are first summed over workloads within an arm and seed; the two seed rates are differenced and then averaged over seeds. This avoids averaging unstable episode ratios and prevents high-opportunity seeds from dominating the algorithm-level estimand.

Uncertainty uses 5,000 crossed bootstrap draws. Each draw independently resamples the eight seed rows and 50 workload columns; ratio endpoints are reconstructed from resampled counts. Exact two-sided sign-flip sensitivity tests enumerate all 2^8 assignments of seed-effect signs. Holm adjustment controls the familywise error rate across delivery and rate effects in two scenarios. The zero-effect test is supplementary; delivery acceptance uses the prespecified non-inferiority bound.

D. Prospective intersection rule

The paper claim is permitted only when all four gates pass in both scenarios: (1) the one-sided 95% lower bound for delivery effect is at least -0.02; (2) the one-sided upper bound for rate effect is below zero; (3) the one-sided upper bound for the constrained absolute rate is at most 0.12; and (4) every constrained policy-seed rate is at most 0.12. A favorable classical comparison cannot compensate for failure.

E. Integrity and failure visibility

The authorization hashes the protocol, runner, MAPPO validation freeze, classical gate freeze, every selected checkpoint, and every Q table. Missing, duplicate, substituted, or non-finite cells fail closed. The entire panel runs regardless of interim outcomes; no seed, scenario, method, or checkpoint can be removed after authorization.

Two pre-test classical executions completed computation but exposed validators that depended on JSON object insertion order. Both executions stopped before any sealed workload was opened. The validators were amended to compare exact job-ID sets while preserving entry-level hashes, the failed directories were retained, and a clean symmetric run produced the bound classical freeze. This operational history is reported because silently replacing a failed run would weaken the study's audit trail.

F. Reproducibility scope

All 48 MAPPO jobs reached the budget; 480 checkpoint-selection recheck rows and 480 independent-gate rows were complete before test authorization. The test runner records row, statistics, and final-freeze hashes. The final sealed artifact contains all 4,100 expected rows with no duplicate key. These controls support exact provenance inside the current code base; they do not substitute for external replication on another simulator.

VI. Sealed-Test Results

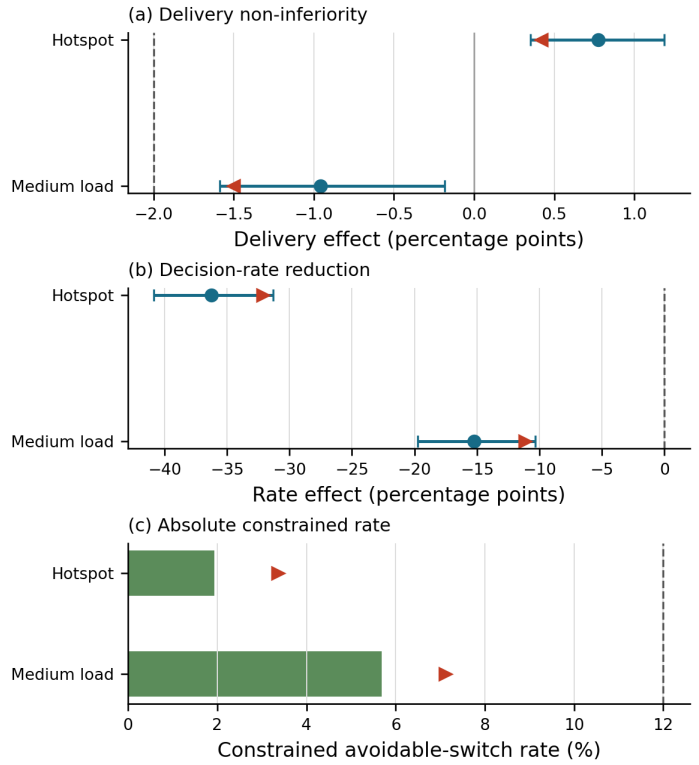


Fig. 2. Primary sealed effects. Circles are point estimates, bars are two-sided 95% crossed-bootstrap intervals, and triangles are one-sided gate bounds. Dashed lines mark the prospective thresholds.

Metric	Medium	Hotspot
Baseline delivery	0.7925	0.2868
Constrained delivery	0.7829	0.2945
Delivery effect, pp	-0.958	+0.772
Delivery 95% CI, pp	[-1.587,-0.182]	[0.353,1.188]
One-sided lower, pp	-1.503	+0.419
Baseline rate	0.2090	0.3818
Constrained rate	0.0568	0.0194
Rate effect	-0.1522	-0.3625
Rate 95% CI	[-.1974,-.1036]	[-.4087,-.3130]
Rate one-sided upper	-0.1114	-0.3214
Maximum seed rate	0.1091	0.0711
Joint gate	PASS	PASS

Table IV. Preregistered primary results.

A. Medium load

The constrained policy reduced the decision-level rate from 0.2090 to 0.0568, an absolute effect of -0.1522 (95% CI -0.1974 to -0.1036). The one-sided rate-effect upper bound was -0.1114, and the absolute-rate upper bound was 0.0712. All eight constrained seed rates were below 0.12; the maximum was 0.1091.

Delivery changed from 0.7925 to 0.7829, or -0.958 percentage points (95% CI -1.587 to -0.182). This is a measurable cost, not an improvement. The one-sided lower bound was -1.503 points and therefore remained above the -2-point non-inferiority margin. The delivery zero-effect sensitivity test had raw and Holm-adjusted $p=0.0546875$; the prospective conclusion comes from the non-inferiority bound, not this superiority-style test.

B. Hotspot-high load

The baseline rate was 0.3818 and the constrained rate 0.0194. Their difference was -0.3625 (95% CI -0.4087 to -0.3130), with a one-sided upper bound of -0.3214 . The constrained absolute-rate upper bound was 0.0336 and the maximum seed rate was 0.0711. Delivery increased from 0.2868 to 0.2945, an effect of $+0.772$ points (95% CI $+0.353$ to $+1.188$); its one-sided lower bound was $+0.419$ points.

The decision-rate effect was negative for all eight policy seeds in both scenarios. Its exact raw p-value was 0.0078125 in each scenario and Holm-adjusted $p=0.03125$. Hotspot delivery favored the constrained arm for seven of eight seeds and had raw $p=0.015625$ and adjusted $p=0.03125$.

Method	Med D/R	Hot D/R
MAPPO QoS	0.7925/0.2090	0.2868/0.3818
MAPPO constrained	0.7829/0.0568	0.2945/0.0194
MAPPO shaped	0.7925/0.1463	0.2885/0.2427
Q-routing	0.7875/0.2605	0.3114/0.3960
OSPF-ECMP	0.7386/0.1875	0.3012/0.1982
Dijkstra	0.7381/0.1886	0.3011/0.1985

Table V. All methods: delivery / decision-level avoidable-switch rate. Classical comparisons are secondary.

C. Reward shaping and classical context

Reward shaping preserved baseline delivery under medium load but reduced the rate only from 0.2090 to 0.1463. Under hotspot load it reached 0.2885 delivery and 0.2427 rate. The explicit constraint was therefore materially more selective in these runs. Because the shaped reward counts accepted physical switches rather than the same pre-contention endpoint, this comparison does not prove that dual optimization dominates every possible fixed coefficient.

Under medium load, constrained MAPPO delivery was close to Q-routing (0.7829 versus 0.7875) and exceeded OSPF-ECMP and Dijkstra, but its switch rate was much lower than all three. Under hotspot load, Q-routing had the highest delivery, 0.3114, while OSPF and Dijkstra were about 0.301 and constrained MAPPO was 0.2945. The method therefore does not dominate the classical baselines on delivery. Its distinguishing result is stability relative to its matched MAPPO reference while meeting the registered delivery tolerance.

D. Seed heterogeneity and gate margins

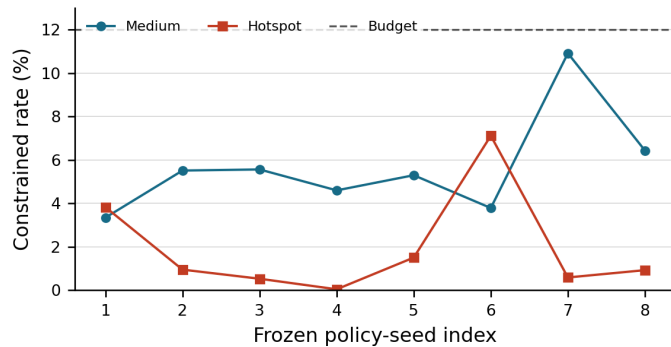


Fig. 3. Constrained decision-level rates for the eight frozen policy seeds. The dashed line is the 12% per-seed budget.

Constrained seed rates ranged from 3.35% to 10.91% under medium load and from 0.04% to 7.11% under hotspot load. The medium panel was the tighter setting: one seed approached

the budget and its weakest paired rate reduction was 2.40 percentage points. The other seven medium effects and all hotspot effects were larger. Reporting this spread matters because the equal-seed mean alone would hide deployment-to-deployment variability.

Table VI expresses each successful gate as signed headroom from its failure boundary. The smallest margin is the medium delivery lower bound, only 0.497 percentage points above the non-inferiority threshold. The next smallest is the medium maximum-seed rate, 1.087 points below budget. Thus, the main conclusion should emphasize passage of a prespecified tolerance rather than claim delivery equivalence or a universal safety guarantee.

Passing margin	Medium	Hotspot
Delivery lower vs -2 pp	+0.497 pp	+2.419 pp
Rate-effect upper vs 0	11.139 pp	32.143 pp
Rate upper vs 12%	4.885 pp	8.635 pp
Max seed vs 12%	1.087 pp	4.886 pp

Table VI. Distance from each prospective failure boundary; positive values indicate passage.

VII. Discussion

The effect depends on load. In the medium regime, the controller spends about one delivered packet per hundred generated packets to remove most avoidable changes. In the hotspot regime, suppressing unstable decisions also improves delivery relative to unconstrained MAPPO, plausibly because concentrated traffic makes local oscillation more damaging. This mechanism is an interpretation, not a separately randomized causal result.

An explicit rate budget improves operational interpretability. The same 0.12 target governs checkpoint feasibility, dual pressure, uncertainty on the test rate, and per-seed compliance. In contrast, a scalar reward coefficient has no invariant meaning when QoS reward scales or traffic mixtures change. The projected update is also simple enough to reproduce: one rate, one multiplier, one update per completed rollout.

The result should not be read as a hard safety guarantee. Stochastic optimization can miss the target, and deployment traffic can move outside the training distribution. The protocol addresses empirical reliability by requiring both a confidence bound and all eight observed seed rates to pass, but it does not establish concentration bounds for a future constellation.

VIII. Limitations and Threats to Validity

A. External validity

Evidence covers only two synthetic regimes in a 24-satellite, 30-slot environment. Orbital geometry is simplified; propagation delay, contact prediction errors, inter-satellite link acquisition, routing-table dissemination, and handover signaling are not modeled at protocol fidelity. A TLE/SGP4 or Hypatia/ns-3 replication is required before making claims about operational constellations, longer horizons, or larger node counts.

B. Observation and overhead assumptions

Execution assumes fresh synchronous one-hop queue, utilization, reliability, and contention telemetry. The simulator includes a soft control-overhead reward component, but this is not a measured control-plane bandwidth result. Delayed, lost, or quantized telemetry could change both candidate ranking and the apparent value of retaining a cached next hop.

C. Endpoint interpretation

A lower local decision-switch rate is a proxy for forwarding stability, not a direct measurement of convergence time, route-flap propagation, or table-update volume. Cache keys and local decisions may map differently to real router state. Conversely, forced changes are intentionally excluded, so the endpoint cannot describe resilience to link disappearance by itself.

D. Statistical scope

There are eight independent policy seeds. Pairing 50 workloads improves precision for the audited conditions but does not create 400 independent learned policies. Crossed bootstrap intervals rely on exchangeability of the registered seed and workload samples. Classical comparisons are secondary, and Dijkstra has only one routing identity; no inferential claim is made for that deterministic method.

E. Engineering maturity

The implementation has extensive integrity checks and exact-resume tests, yet it remains a research simulator. Wall-clock inference latency, accelerator dependence, memory use, routing message integration, and failure recovery on flight-like hardware have not been measured. The current artifact is adequate for the bounded algorithmic claim, not for readiness or deployment claims.

IX. Reproducibility and Claim Boundary

The MAPPO evidence consists of 48 completed 50k-step jobs, frozen checkpoints, and complete selection and gate panels. The classical freeze contains 16 trained Q-routing models and 340 independent gate rows. The one-time sealed panel contains every registered method-scenario-seed-workload cell. Source and artifacts are bound by SHA-256 manifests; the final sealed freeze is c5cc82c7edb96add6e8da1275924e83cf159e65a2062184d52f3364ad8e508a9.

Frozen stage	Complete evidence
MAPPO training	48 jobs; 50,040 steps/job
Selection recheck	480 paired rows
Independent gate	480 paired rows
Classical training	16 Q-routing tables
Classical gate	340 rows
Joint sealed test	4,100 rows
Grid integrity	0 missing; 0 duplicate

Table VII. Frozen evidence coverage used by the submission claim.

The supported claim is consequently narrow: in the audited simulator, under medium and hotspot-high load, this explicit constraint substantially reduced decision-level avoidable switching relative to matched QoS-only MAPPO while meeting a prospective delivery non-inferiority rule. The study does not establish throughput optimality, global routing optimality, universal baseline superiority, zero signaling overhead, or generalization beyond the tested scenarios.

X. Conclusion

Constraint-aware candidate-set MAPPO reduced avoidable cached-next-hop changes by 15.22 and 36.25 percentage points in a sealed two-scenario evaluation. Delivery paid a 0.96-point cost under medium load and gained 0.77 points under hotspot load; both passed the registered two-point non-inferiority margin. The actor retains masked, permutation-equivariant decentralized execution, while the graph critic and projected multiplier remain training-only. These results justify further evaluation on realistic constellation traces and protocol-level simulators, with the current sealed result preserved unchanged as the initial controlled evidence.

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